Aryan Mishra

AI/ML Engineer | Final-Year CS (Data Science) Undergraduate
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Summary

Final-year Computer Science (Data Science) undergraduate and aspiring AI/ML Engineer with hands-on experience designing and shipping 3 end-to-end AI systems: a RAG evaluation harness, a multi-agent healthcare platform, and a multilingual NLP engine. Core strengths in Python, FastAPI, LangGraph, RAG pipelines, and MLOps tooling (Docker, CI/CD, MLflow). Track record of measurable results: 100% top-5 hit rate retrieval, 92% agent-routing precision, 78.1% multilingual sentiment F1 – validated with automated test suites and CI quality gates.

Skills

Programming Languages: Python, SQL, Bash
AI/ML: PyTorch, scikit-learn, LangChain, LangGraph, Hugging Face Transformers, Natural Language Processing (NLP), Prompt Engineering, RAG, Multi-Agent Systems, Model Quantization (ONNX INT8)
Retrieval & Search: ChromaDB, FAISS, BM25, Vector Databases, Hybrid Retrieval, Semantic Search
Backend: FastAPI, REST APIs, PostgreSQL, SQLAlchemy, Pydantic, JWT Auth
Cloud & MLOps: Google Cloud Platform (GCP), Docker, GitHub Actions (CI/CD), MLflow, DVC, Git, pytest
Protocols: Model Context Protocol (MCP)

Personal Projects

RAG EvalForge (Personal Project) – Production-Grade RAG Evaluation Harness *April 2026 – August 2026*
Code repository: github.com/Aryanmishra-dev

- Benchmarked 4 chunking strategies (fixed, recursive, sentence, semantic) on a hand-labeled QA dataset, reaching 100% top-5 hit rate and 0.858 MRR with the recursive strategy.
- Diagnosed semantic-chunking over-segmentation via threshold sweeps, improving top-5 retrieval accuracy by 22%.
- Combined dense (ChromaDB), lexical (BM25),and cross-encoder re-ranking with Reciprocal Rank Fusion, cutting retrieval latency by 35%.
- Built an LLM-as-a-judge evaluator scoring faithfulness, correctness, and relevance across embedding models.
- Enforced a CI quality gate (108 pytest cases, 89% coverage, pyright strict, bandit clean) on every PR.
- Exposed an MCP server for ingestion, benchmarking, and registry tools, consumable by clients like Claude Desktop.

MedAgent AI (Personal Project) – Multi-Agent Healthcare Conversational Platform *November 2025 – March 2026*
Code repository: github.com/Aryanmishra-dev

- Designed a LangGraph supervisor routing queries across 4 specialized agents (Research, Chat, Recommendation, Summarization), reaching 92% intent-routing precision on 500+ test scenarios.
- Built a domain RAG pipeline over a healthcare knowledge base with persistent memory, improving multi-turn context retention by 30%.
- Implemented JWT auth, SSE token streaming, and Celery + Redis async task processing on a FastAPI/PostgreSQL backend.
- Containerized the stack with Docker Compose and load-tested it for 500+ concurrent sessions behind Nginx.

Multilingual ABSA (Personal Project) – Cross-Lingual Aspect-Based Sentiment Engine *June 2025 – October 2025*
Code repository: github.com/Aryanmishra-dev

- Built a multilingual ABSA engine for English, Hindi and Hinglish reviews, reaching 78.1% F1 (English) and 67.8% F1 (Hindi) with ONNX INT8-quantized models at 185ms p95 latency.
- Designed a fallback inference chain (INT8 to FP32 to rule-based) guaranteeing 100% % request availability.
- Tracked a full ML pipeline (preprocess to train to quantize to export) using DVC and MLflow, including drift monitoring.
- Streamlit shipped a web app that supports single-record and batch-CSV predictions with live monitoring dashboards.

Education

Indore Institute of Science and Technology *Expected May 2027*
Bachelor of Technology in Computer Science Engineering (Data Science) *CGPA: 7.41/10 (74.1%)*
Relevant Coursework: Data Structures, Algorithms, Machine Learning, Deep Learning, Database Systems, Statistics

Certifications

- NVIDIA Deep Learning Institute – Anomaly Detection for Predictive Maintenance and Computer Vision (2025)
- Google Cloud – Generative AI Fundamentals (2026)
- Google Cloud – Large Language Model Implementations (2026)
- Anthropic – Model Context Protocol, Claude 101, Claude Code 101, Introduction to Agent Skills
- Cisco – Apply AI: Analyze Customer Reviews; Intro to Modern AI (2026)

Achievements

- Kaggle Bronze Medalist – Kaggle March Machine Learning Mania 2024